

B7:

$$a) \frac{2}{y} - \frac{1}{5} = \frac{x}{15}$$

$$\frac{2}{y} = \frac{x}{15} + \frac{1}{5}$$

$$\frac{2}{y} = \frac{3+x}{15}$$

$$(3+x) \cdot y = 30$$

$\forall x, y \in \mathbb{Z}$ nên $3+x \in \mathbb{Z}$

$y, 3+x \in U(30)$

Mà $U(30) = \{\pm 1, \pm 2, \pm 3, \pm 5, \pm 6, \pm 10, \pm 15, \pm 30\}$

$$b) \frac{y}{2} + \frac{1}{4} = \frac{5}{x}$$

$$\frac{y}{2} = \frac{5}{x} - \frac{1}{4}$$

$$\frac{5}{x} = \frac{y}{2} + \frac{1}{4}$$

$$\frac{5}{x} = \frac{2y+1}{4}$$

$$\Rightarrow (2y+1) \cdot x = 5 \cdot 4$$

$$(2y+1) \cdot x = 20$$

B8:

$$a) 81^4 \cdot 3^5 = (3^4)^4 \cdot 3^5 = 3^{16} \cdot 3^5 = 3^{21}$$

$$b) 64^2 : 49^3 = (8^2)^2 : (7^2)^3 = (2^3)^2 : (7^2)^3$$

$$= 2^6 : 7^6 = \left(\frac{2}{7}\right)^6$$

$$c) (0,5)^3 \cdot 32 = (0,5)^3 \cdot 2^5 = (0,5 \cdot 2)^5 = 1^5 = 1$$

$$d) \left(\frac{6}{13}\right)^3 \cdot \left(\frac{26}{9}\right)^3 = \left(\frac{6}{13} \cdot \frac{26}{9}\right)^3 = \left(\frac{4}{3}\right)^3$$

$$e) \left(\frac{5}{7}\right)^9 \cdot \frac{25}{49} = \left(\frac{5}{7}\right)^9 \cdot \left(\frac{5}{7}\right)^2 = \left(\frac{5}{7}\right)^{11}$$

1,75 a) $\left(\frac{6}{5}\right)^4 : \frac{36}{25} = \left(\frac{6}{5}\right)^4 : \left(\frac{6}{5}\right)^2 = \left(\frac{6}{5}\right)^2$ ✓

B2:

a) $\frac{14^6}{2^5 \cdot 49^3} = \frac{2^6 \cdot 7^6}{2^5 \cdot (7^2)^3} = \frac{2^6 \cdot 7^6}{2^5 \cdot 7^6} = \frac{2}{1} = 2$ ✓

b) $\frac{6^5 \cdot 5^6}{4^3 \cdot 75^3} = \frac{6^5 \cdot 5^6}{4^3 \cdot 15^3 \cdot 5^3} = \frac{6^5 \cdot 5^3}{4^3 \cdot 5^3 \cdot 3^3} = \frac{6^5}{4^3 \cdot 3^3}$
 $= \frac{3^5 \cdot 2^5}{2^3 \cdot 3^3} = \frac{3^2 \cdot 2^5}{2^3 \cdot 2^3} = \frac{3^2}{2} = \frac{9}{2}$ ✓

c) $\frac{9^2 + 2 \cdot 3^5 + 5 \cdot 729}{3^4} = \frac{81 + 6 + 5 \cdot 9^3}{3^4}$

d) $\frac{9^2 + 2 \cdot 3^5 + 5 \cdot 729}{3^4} = \frac{9^2 + 2 \cdot 3^5 + 5 \cdot 9^3}{3^4}$

$= \frac{(3^2)^2 + 2 \cdot 3^5 + 5 \cdot (3^2)^3}{3^4} = \frac{3^4 + 2 \cdot 3^5 + 5 \cdot 3^6}{3^4}$

$= \frac{3^4 + 2 \cdot 3^5 + 5 \cdot 3^6}{3^4} = \frac{3^4 + 2 \cdot 3^5 + 5 \cdot 3^5}{3^4}$

$= \frac{3^4 (1 + 2 + 5)}{3^4} = 3 \cdot 68 = 24$ 52

e) $\frac{5^{19} + 125^7}{5^{16} \cdot 5^4 + 12 \cdot 5^{19}} = \frac{5^{19} + (5^3)^7}{5^{16} \cdot 5 + 12 \cdot 5^{19}} = \frac{5^{19} + 5^{21}}{5^{19} \cdot (5 + 12)}$

$= \frac{5^{19} + 5^{21}}{5^{19} \cdot 17} = \frac{5^{19} (1 + 5^2)}{5^{19} \cdot 17} = \frac{5^{19} \cdot 26}{5^{19} \cdot 17} = \frac{26}{17} = 2$

B3:

a) $(2x - 3)^3 = 125$

$(2x - 3)^3 = 5^3$

$2x - 3 = 5$

$2x = 8$

$x = 4$
 Vậy $x = 4$ ✓

b) $(0,2x - 1,5)^2 = 6,25$

$\left(\frac{1}{5}x - \frac{3}{2}\right)^2 = \frac{25}{4}$

$\left(\frac{1}{5}x - \frac{3}{2}\right)^2 = \left(\frac{5}{2}\right)^2$

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$$\text{TH1: } \frac{1}{5}x - \frac{3}{2} = \frac{5}{2}$$

$$\frac{1}{5}x = 4$$

$$x = 4 : \frac{1}{5}$$

$$x = 20$$

$$\text{TH2: } \frac{1}{5}x - \frac{3}{2} = -\frac{5}{2}$$

$$\frac{1}{5}x = \frac{-5}{2} + \frac{3}{2}$$

$$\frac{1}{5}x = \textcircled{-1} - 1$$

$$x = \textcircled{-5}$$

$$\text{c) } \left(\frac{1}{7}\right)^{2x+1} = \sqrt[2]{\left(\frac{1}{7}\right)^2} \quad \text{Vây } x \in \{20, 5\}$$

$$\Rightarrow 2x+1 = 2$$

$$2x = 1$$

$$x = \frac{1}{2}$$

$$\textcircled{1,85}$$

$$\text{d) } \left(-\frac{5}{4}\right)^{x+\frac{1}{3}} = \frac{25}{16}$$

$$\left(-\frac{5}{4}\right)^{x+\frac{1}{3}} = \left(\frac{5}{4}\right)^2$$

$$\left(-\frac{5}{4}\right)^{x+\frac{1}{3}} = \left(-\frac{5}{4}\right)^2$$

$$\Rightarrow x + \frac{1}{3} = 2$$

$$x = \frac{6}{3} - \frac{1}{3}$$

$$x = \frac{5}{3}$$

$$\text{e) } (x - 2025)^{19} = (x - 2025)^{20}$$

$$\text{TH1: } 0^{19} = 0^{20} = 0$$

$$\text{Tácó: } x - 2025 = 0 \Rightarrow x = 2025$$

$$\text{Vậy } x = 2025$$

$$1+12: 1^{10} = 1^{20}$$

$$x - 2025 = 1$$

$$\Rightarrow x = 2026. \text{ Suy ra } x \in \{2026; 2025\} \quad \text{P}$$

B4:

$$2) (x-2)^2 + (2y+3)^2 = 0$$

$$(1) \begin{cases} (x-2)^2 \geq 0 \text{ với mọi } x \in \mathbb{Q} \\ (2y+3)^2 \geq 0 \text{ với mọi } y \in \mathbb{Q} \end{cases}$$

$$\text{Từ (1) suy ra } \begin{cases} x-2=0 \\ 2y+3=0 \end{cases}$$

Hay

$$\begin{cases} x=2 \\ y=-\frac{3}{2} \end{cases}$$

$$\text{Vậy } x=2; y=-\frac{3}{2} \quad \text{P}$$

1,25

$$b) (2x-3)^2 + (y^2-9)^4 \leq 0$$

$$\text{Vì } (2x-3)^2 \geq 0 \text{ vs mọi } x$$

$$(y^2-9)^4 \geq 0 \text{ vs mọi } x$$

$$\text{Nên } (2x-3)^2 + (y^2-9)^4 \geq 0 \text{ vs mọi } x, y \in \mathbb{Q}$$

$$\text{Suy ra } (2x-3)^2 + (y^2-9)^4 = 0$$

$$\text{Vậy } (x; y) \in \left\{ \left(\frac{3}{2}; 3 \right); \left(\frac{3}{2}; -3 \right) \right\} \quad \text{P}$$

B5:

$$2) 25^9 \text{ và } 125^6$$

$$\Rightarrow 25^9 \text{ và } 125^6$$

$$\Rightarrow (5^2)^9 \text{ và } (5^3)^6$$

$$\Rightarrow 5^{18} > 5^9$$

$$\text{Vậy } 25^9 > 125^6 \quad \text{P}$$

$$b) 2^{30} \text{ và } 3^{20}$$

$$\Rightarrow (2^3)^{10} \text{ và } (3^2)^{10}$$

$$\Rightarrow 8^{10} \text{ và } 9^{10}$$

$$\Rightarrow 8^{10} < 9^{10}$$

$$\text{Vậy } 2^{30} < 3^{20} \quad \text{P}$$

c) $\left(-\frac{1}{27}\right)^{150}$ và $\left(-\frac{1}{9}\right)^{200}$

Ta có $\left(-\frac{1}{27}\right)^{150} = \left(\frac{1}{27}\right)^{150} = \left[\left(\frac{1}{3}\right)^3\right]^{150} = \left(\frac{1}{3}\right)^{450}$
 $\left(-\frac{1}{9}\right)^{200} = \left(\frac{1}{9}\right)^{200} = \left(\frac{1}{3}\right)^{400}$
 $= \left(\frac{1}{3}\right)^{400}$

Ta có $\left(\frac{1}{3}\right)^{450}$ và $\left(\frac{1}{3}\right)^{400}$; $450 > 400$

Vậy $\left(\frac{1}{3}\right)^{450} > \left(\frac{1}{3}\right)^{400}$ Hay $\left(-\frac{1}{27}\right)^{150} < \left(-\frac{1}{9}\right)^{200}$

d) $\left(-\frac{27}{125}\right)^{30}$ và $\left(-\frac{243}{3125}\right)^{18}$

Ta có: $\left(-\frac{27}{125}\right)^{30} = \left(\frac{27}{125}\right)^{30} = \left[\left(\frac{3}{5}\right)^3\right]^{30} = \left(\frac{3}{5}\right)^{90}$
 $\left(-\frac{243}{3125}\right)^{18} = \left[\left(\frac{3}{5}\right)^5\right]^{18} = \left(\frac{3}{5}\right)^{90}$
 $\Rightarrow \left(\frac{3}{5}\right)^{90} = \left(\frac{3}{5}\right)^{90}$

Suy ra $\left(-\frac{27}{125}\right)^{30} = \left(-\frac{243}{3125}\right)^{18}$

B_b:

1) $A = 5(x+3)^2 + 1$

Do $(x+3)^2 \geq 0$ với mọi $x \in \mathbb{Q}$

Nên $5(x+3)^2 \geq 0$

Suy ra $5(x+3)^2 + 1 \geq 1$

Hay $A \geq 1$. Dấu "=" xảy ra khi $x+3=0$

Hay $x = -3$

Vậy GTNN của $A = 1$ khi $x = -3$

$$b) B = (2x-1)^2 + 2(5-2y)^2 + 2026$$

Ta có $(2x-1)^2 \geq 0$ và $2(5-2y)^2 \geq 0$ với mọi x, y

Suy ra $B \geq 2026$

Dấu "=" xảy ra khi $\begin{cases} 2x-1=0 \\ 2(5-2y)^2=0 \end{cases}$
 Hay $\begin{cases} 2x-1=0 \\ 5-2y=0 \end{cases}$ hay $\begin{cases} x=\frac{1}{2} \\ y=\frac{5}{2} \end{cases}$

Vậy GTNN của B là 2026 khi $x=\frac{1}{2}; y=\frac{5}{2}$

B7:

$$1^2 + 2^2 + 3^2 + \dots + 10^2 = 385$$

$$T = 12^2 - 1^2 + 14^2 - 3^2 + 16^2 - 5^2 + 18^2 - 7^2 + 20^2 - 9^2$$

$$T = (12^2 + 14^2 + 16^2 + 18^2 + 20^2) - (1^2 + 3^2 + 5^2 + 7^2 + 9^2)$$

$$T = (12^2 + 14^2 + 16^2 + 18^2 + 20^2) + (2^2 + 4^2 + 6^2 + 8^2 + 10^2) - (2^2 + 4^2 + 6^2 + 8^2 + 10^2) - (1^2 + 3^2 + 5^2 + 7^2 + 9^2)$$

$$= (2^2 + 4^2 + 6^2 + 8^2 + \dots + 18^2 + 20^2) - (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2)$$

$$= (2^2 + 2^2 \cdot 2^2 + 2^2 \cdot 3^2 + \dots + 2^2 \cdot 9^2 + 2^2 \cdot 10^2) - 385$$

$$= 2^2 \cdot (1^2 + 2^2 + 3^2 + \dots + 9^2 + 10^2) - 385$$

$$= 2 \cdot 385 - 385 = 385$$

9, 6 đ

7 (7b)

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- Cô Phen Minh Khuê có ý thức làm BSTVN

- Con cần chú ý:

+) Tính toán, đổi dấu cẩn thận hơn

- Về con làm lại bài sai vào vở

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